JINAL SUTHAR

AI/ML Engineer

+91-8005646737 • jinaaa1115@gmail.com • Ahmedabad, Gujarat

linkedin.com/in/jinaaaa110105 • https://github.com/Jinaaa1115

PROFESSIONAL SUMMARY

Aspiring AI/ML Engineer currently completing a Data Science with AI program at TOPS Technologies after completing a Bachelor of Computer Applications from JNVU. Proficient in Python, SQL, Machine Learning, Deep Learning, FastAPI, Docker, Git, and GitHub Actions, with hands-on use of ChatGPT and Claude to accelerate coding, debugging, SQL development, and AI application prototyping. Built and deployed end-to-end machine learning applications, including predictive models and a customer segmentation system analysing over 500K transactions, and targeting AI/ML Engineer or Data Scientist roles.

TECHNICAL SKILLS

Languages & Querying: *Python, SQL (MySQL), HTML, CSS, JavaScript*

Data Analysis & ML Libraries: *pandas, NumPy, scikit-learn, Matplotlib, Seaborn, TensorFlow*

Frameworks & Technologies: *FastAPI, Flask, REST APIs, LangChain, LangGraph, CrewAI, MCP*

DevOps & Tools: *Git, GitHub, GitHub Actions, Docker*

AI Tools & Platforms: *ChatGPT, Claude, Google Gemini*

Databases & Cloud: *MySQL, MongoDB, Aiven (Managed MySQL), Render (Hosting)*

PROJECTS

Project 1 | *AI Business Analyst Agent*

Tech Stack: *Python (pandas, NumPy), FastAPI, LangChain, Google Gemini, React, MySQL, Docker-free deploy on Render, GitHub Actions*

Dataset: *User-uploaded CSV files (dynamic, any tabular business dataset)*

Live Link: *ai-business-analyst-agent*

GitHub: *https://github.com/Jinaaa1115/AI_Business_Analyst_Agent.git*

Summary:

Developed an end-to-end AI agent application that lets users upload CSV datasets and query them in natural language, returning grounded findings and recommendations without hallucinated numbers.

- *Designed a LangChain tool-calling agent powered by Google Gemini with six deterministic pandas-based analysis tools (dataset profiling, aggregation, statistics, trend analysis, comparison, visualization), ensuring all numeric outputs are computed rather than generated.*
- *Engineered persistent chat history and dataset storage using FastAPI, SQLAlchemy, and MySQL, including content-hash-based duplicate detection and automatic recovery from server restarts via database-backed file storage.*
- *Built a responsive React (Vite + Tailwind) frontend with inline Plotly chart rendering, markdown-formatted AI responses, and a dataset/chat history dashboard; set up CI/CD with GitHub Actions to auto-deploy backend and frontend to Render on every push.*

- **AI Integration:** *Used **ChatGPT and Claude** to debug LangChain agent architecture, resolve model/API versioning issues, optimize SQLAlchemy relationships, and design the tool-calling system — significantly accelerating development and deployment.*

Project 2 | ***Customer Segmentation Engine***

Tech Stack: *Python (pandas, NumPy, scikit-learn), FastAPI, React, Docker, GitHub Actions, REST API*

Dataset: *Online Retail II Dataset (UCI/Kaggle), **500K+ customer transactions***

Live Link: *customer-segmentation-engine*

GitHub: *https://github.com/Jinaaa1115/customer_segmentation.git*

Summary:

Developed an end-to-end customer segmentation application that analyses customer purchasing behaviour using RFM analysis and K-Means clustering, enabling businesses to identify high-value customer groups for targeted marketing campaigns

- ***Engineered** Recency, Frequency, and Monetary (RFM) features from **500K+ retail transactions**, performed preprocessing, feature scaling, and clustering to create an analysis-ready customer dataset.*
- ***Built and evaluated** a K-Means clustering model that segmented customers into **4 behavioural groups**, identifying a **'Champion' segment comprising approximately 13% of customers** with significantly higher average spending.*
- **AI Integration:** *Used **ChatGPT and Claude** to accelerate feature engineering, debug clustering workflows, optimize FastAPI endpoints, and streamline Docker configuration, reducing development time while improving deployment reliability.*

Project 3 | ***House Price Prediction Platform***

Tech Stack: *Python (pandas, NumPy, scikit-learn), Flask, React, Docker, GitHub Actions, REST API*

Dataset: *House Price Dataset (Kaggle), housing attributes including area, bedrooms, bathrooms, and location features*

Live Link: *house-price-prediction*

GitHub-backend-repo: *https://github.com/Jinaaa1115/house_prediction_backend.git*

Summary:

Developed and deployed a full-stack machine learning application that predicts residential property prices based on user-provided housing features through a REST API and React frontend.

- ***Preprocessed** housing data by handling missing values, encoding categorical variables, and engineering features to train regression models for house price prediction.*
- ***Built and deployed** an end-to-end prediction pipeline using **Flask REST API**, React frontend, Docker, and GitHub Actions, enabling users to generate real-time house price estimates through a web interface.*
- **AI Integration:** *Leveraged **ChatGPT and Claude** to accelerate model experimentation, optimize preprocessing pipelines, debug backend APIs, and assist in frontend integration, significantly improving development efficiency.*

EDUCATION

Bachelor of Computer Applications (BCA)

JNVU

2025 • CGPA: 8.42/10

Certification — Data Science with AI

TOPS Technologies, Ahmedabad

Dec 2025 – Present • Key Modules: SQL, Python, Pandas, NumPy, Scikit-learn, Machine Learning, Deep Learning, Model Deployment, REST API Development, Natural Language Processing (NLP), Large Language Models (LLMs), Computer Vision, and Agentic AI

LANGUAGES

- *English*
- *Hindi*

CERTIFICATIONS

- ***Data Science with AI** — TOPS Technologies (In Progress)*